

Lab 11: Faraday's Law

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Pre-lab Questions

1. The first graph shows that the magnetic flux has maximum magnitude when the center of the magnet is at the center of the loop. What is the induced emf at this instant?

The induced emf when the flux reaches its maximum is 0 V. Since the magnetic flux reaches an extremum at this instant, its derivative with respect to time is zero. By Faraday's law, the induced emf is equal to the negative rate of change of magnetic flux, so the emf is zero at this instant. Alternatively, we can reason that at this instant, the magnetic flux through the loop goes from increasing to decreasing in magnitude, so there is a point at which the flux is not changing at all, which is where the emf is zero.

2. In the simulation, positive flux corresponds to field lines passing from left to right through the loop. Would either of the graphs change if the magnet was turned around and the south pole passed through the loop first?

The graphs would be flipped. Magnetic field lines on a bar magnet emerge from the north pole and enter the south pole. When the north pole passes through the loop from right to left, the magnetic field emerging from the north pole points from right to left, causing a negative magnetic flux. If we flip the magnet, the south pole would pass through the loop first, and the magnetic field lines would point from left to right, causing a positive magnetic flux. The emf graph would also flip because the emf is proportional to the rate of change of flux.

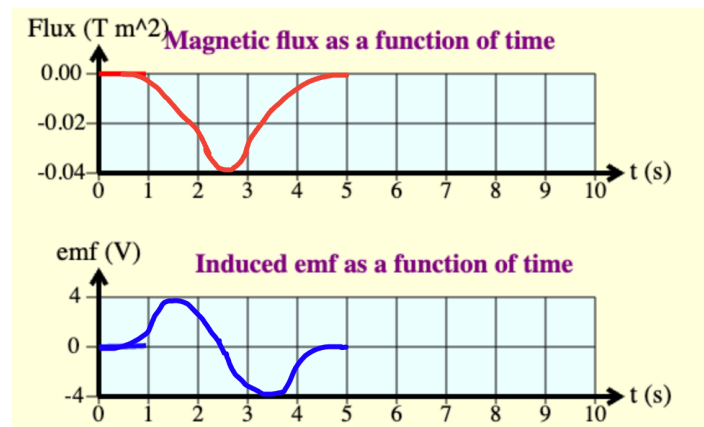
3. If the velocity of the magnet is doubled, would the graph of magnetic flux change?

The magnetic flux graph would be "compressed" horizontally by a factor of two. That is, the time at which we reach maximum magnetic flux magnitude will be 2.5 s instead of 5 s, and the width of the peak will be half as wide. However, the height of the peak will not change.

4. Would the graph of induced emf change?

The emf graph would also be compressed horizontally by a factor of two, but it would also be stretched vertically by a factor of two. This is because the emf is proportional to the rate of change of flux, so doubling the velocity doubles the slope of the flux graph at every point, doubling the emf.

Sketch of graphs for answers to questions 3 and 4:



Part 1: Time-dependent Magnetic Flux from Permanent Magnet Dropped through a Coil of Wire

Data Table

Magnet	Pole Down	Drop Height	Peak 1			Peak 2		
			+ / -	$ \mathcal{E} _{\max}$	Area	+ / -	$ \mathcal{E} _{\max}$	Area
Strong	<i>N</i>	2 cm	+	3.454 V	0.2630 Wb	-	5.508 V	0.2674 Wb
Strong	<i>S</i>	2 cm	-	3.207 V	0.2645 Wb	+	5.392 V	0.2664 Wb
Strong	<i>N</i>	10 cm	+	4.667 V	0.2670 Wb	-	6.250 V	0.2662 Wb
Weak	<i>N</i>	2 cm	+	1.168 V	0.0574 Wb	-	1.461 V	0.0578 Wb
Weak	<i>S</i>	2 cm	-	1.045 V	0.0693 Wb	+	1.522 V	0.0689 Wb
Weak	<i>N</i>	10 cm	+	2.325 V	0.0677 Wb	-	1.485 V	0.0702 Wb

Questions

1. When the magnet is dropped through the coil with the south pole down, why do we see two peaks and why are the two peaks in opposite directions?

When the magnet enters the coil, the magnetic flux through the coil increases, so the induced emf opposes this increase in flux, resulting in a negative peak. When the magnet leaves the coil, the magnetic flux through the coil decreases, so the induced emf opposes this decrease in flux, resulting in a positive peak. The two peaks are in opposite directions because they correspond to opposite changes in magnetic flux.

2. How do the two peaks change if the magnet is dropped with the north pole down?

The two peaks flip, since the magnetic field from the magnet would point in the opposite direction, flipping the sign of the magnetic flux and thus the induced emf.

3. Why is the second peak higher than the incoming peak?

Gravity causes the magnet to accelerate as it falls, so the magnet is moving faster when it leaves the coil than when it enters the coil. Since the emf is proportional to the *rate of change* of flux, and the flux changes more rapidly when the magnet is moving faster, the second peak is higher than the first peak.

4. What does the area under the curve Potential vs. Time represent?

Faraday's law can be rearranged as follows:

$$\mathcal{E} = N \frac{d\Phi_B}{dt} \rightarrow \mathcal{E} dt = N d\Phi_B \rightarrow \int \mathcal{E} dt = N \Delta\Phi_B.$$

The area under a potential vs. time graph is the net change in magnetic flux through the coil as the magnet passes through the coil, times the number of turns in the coil.

5. Are the two areas the same? Why or why not?

The areas enclosed by the two peaks are approximately the same, meaning the net change in magnetic flux was zero. This was reflected in our experiment, which makes sense because the magnet is not inside our coil before and after every trial, so there should be no change in magnetic flux.

6. *How does the magnitude of the induced EMF depend on the speed of the magnet?*

The magnitude of the induced emf is directly proportional to the speed of the magnet.

7. *How does the magnitude of the induced EMF depend on the strength of the magnet?*

The magnitude of the induced emf is directly proportional to the strength of the magnet.

8. *How would the magnitude of the induced EMF change if two magnets were taped together so that:*

- a. *both south ends were together?*

Since the magnets face each other in opposite directions, their magnetic fields will cancel each other out. This results in a much weaker magnetic field, which means the magnitude of the induced emf will be much smaller.

- b. *south end of one magnet is taped to north end of the other?*

Since the magnets face each other in the same direction, their magnetic fields will reinforce each other. This results in a much stronger magnetic field, which means the magnitude of the induced emf will be much larger.

9. *Would the experiment give similar results if the magnet would be stationary and the coil would be dropped?*

Yes, the experiment would give similar results. The relative motion between the magnet and the coil is what causes the change in magnetic flux, so it does not matter whether the magnet is moving and the coil is stationary or vice versa.

10. *Using your data and the known geometries, estimate the strength of the magnetic field of the two bar magnets? Assume magnetic field is confined to the magnet. What extra measurements do you need? Compare your estimate with an actual measurement of the magnetic field using the magnetic field sensor.*

From the derivation in (4), the magnetic flux through the coil changes from approximately zero when the magnet is outside the coil to $N\Phi_B = \int \mathcal{E} dt$ when the magnet is fully inside the coil. So:

$$\Phi_B = \frac{1}{N} \int \mathcal{E} dt.$$

That is, the maximum magnetic flux through the coil is approximated by one of the two peak areas divided by the number of turns in the coil.

Since the magnetic field is approximately uniform across and perpendicular to the area of the coil, we can assume that $\Phi_B \approx BA$:

$$\Phi_B \approx BA = \frac{1}{N} \int \mathcal{E} dt \rightarrow B = \frac{1}{NA} \int \mathcal{E} dt = \frac{\text{area enclosed by one peak}}{NA}.$$

From the additional data in the instructions, we have $N = 3300$ and $A = 4.155 \cdot 10^{-4} \text{ m}^2$, so let $\alpha = 1/(NA) = 0.7294 \text{ m}^{-2}$.

Then, $B = \alpha \cdot \text{area enclosed by one peak}$. The average of the six peak areas from the strong magnet is approximately 0.2658 Wb, so the estimated magnetic field strength of the strong magnet is:

$$B_{\text{strong}} \approx \alpha \cdot 0.2658 \text{ Wb} = 0.1938 \text{ T}.$$

The average of the six peak areas from the weak magnet is approximately 0.06522 Wb, so the estimated magnetic field strength of the weak magnet is:

$$B_{\text{weak}} \approx \alpha \cdot 0.06522 \text{ Wb} = 0.04757 \text{ T}.$$

Note: We were unable to measure the actual strengths of the magnets.

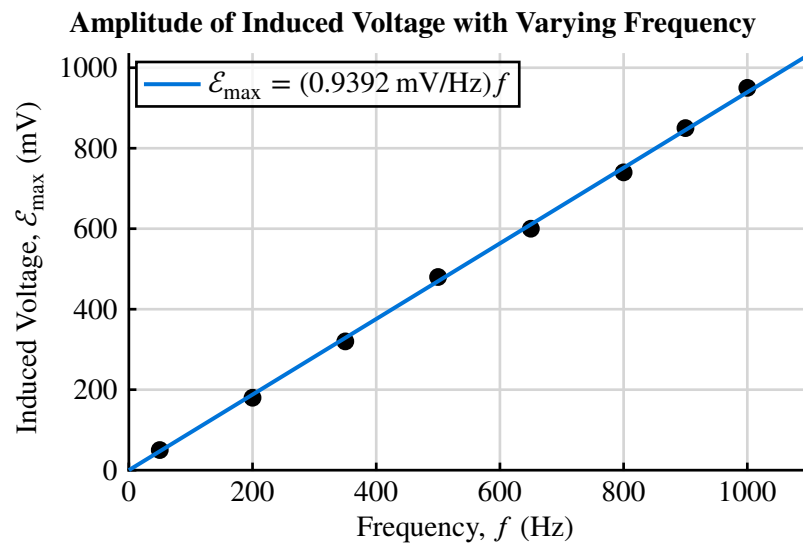
Part 2: Primary and Secondary Coils

Induced Voltage, Primary Voltage Fixed

Without Iron Core ($\mu \approx \mu_0$)

Amplitude of Primary Voltage Signal	$V_0 = 40 \text{ mV}$
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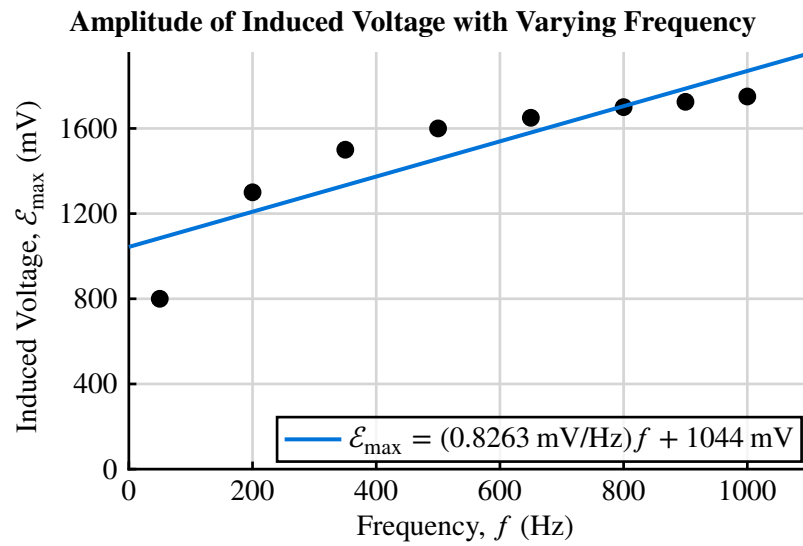
Frequency f	Induced Voltage $\mathcal{E}_{\text{max}}$
50 Hz	50.0 mV
200 Hz	180 mV
400 Hz	320 mV
500 Hz	480 mV
700 Hz	600 mV
800 Hz	740 mV
900 Hz	850 mV
1000 Hz	950 mV



With Iron Core ($\mu \gg \mu_0$)

Amplitude of Primary Voltage Signal	$V_0 = 190 \text{ mV}$
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Frequency f	Induced Voltage $\mathcal{E}_{\text{max}}$
50 Hz	800 mV
200 Hz	1300 mV
400 Hz	1500 mV
500 Hz	1600 mV
700 Hz	1650 mV
800 Hz	1700 mV
900 Hz	1730 mV
1000 Hz	1750 mV



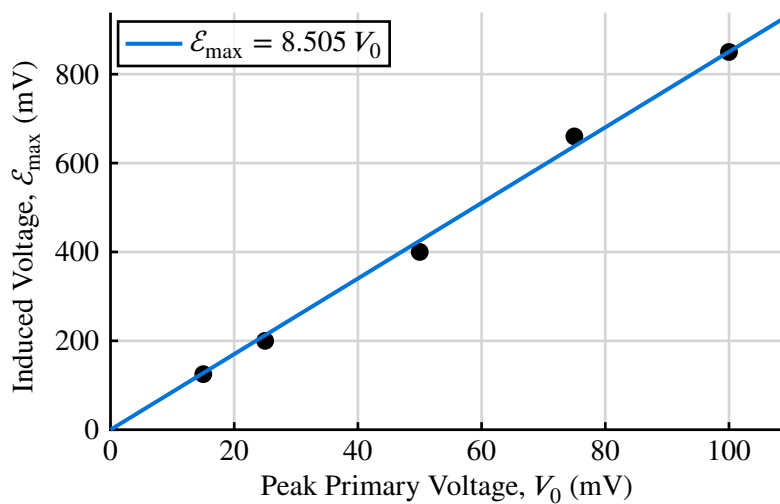
Induced Voltage, Frequency Fixed

Frequency	$f = 500 \text{ Hz}$
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Without Iron Core ($\mu \approx \mu_0$)

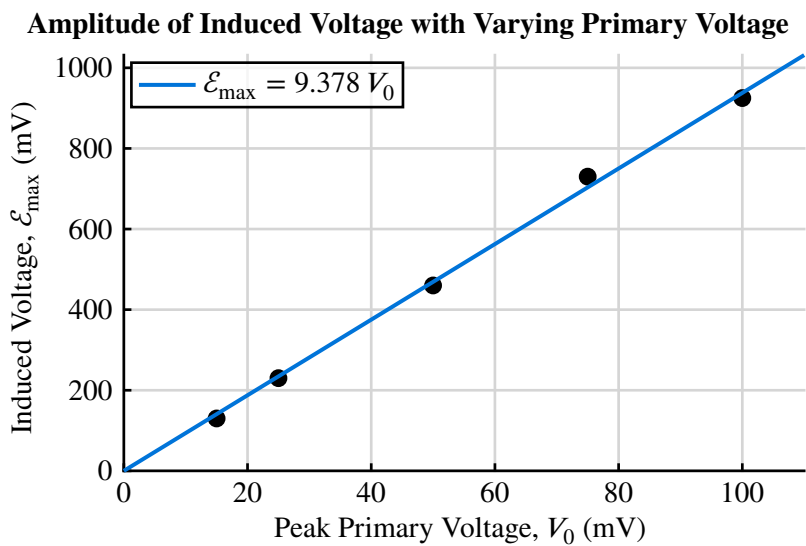
Peak Primary Voltage V_0	Induced Voltage $\mathcal{E}_{\text{max}}$
15.0 mV	125 mV
25.0 mV	200 mV
50.0 mV	400 mV
75.0 mV	660 mV
100 mV	850 mV

Amplitude of Induced Voltage with Varying Primary Voltage



With Iron Core ($\mu \gg \mu_0$)

Peak Primary Voltage V_0	Induced Voltage $\mathcal{E}_{\max}$
15.0 mV	130 mV
25.0 mV	230 mV
50.0 mV	460 mV
75.0 mV	730 mV
100 mV	925 mV



Questions

1. Derive an expression for the magnetic flux in the primary/inner coil.

Let f be the frequency of the function generator, so $\omega = 2\pi f$ is the angular frequency of the potential. Let V_0 be the amplitude of the potential created by the function generator.

The function generator creates a potential $V(t)$ which follows:

$$V(t) = V_0 \sin(\omega t) = V_0 \sin(2\pi f t).$$

Let the primary solenoid have turn density n_1 , resistance R , and cross-sectional area A . To account for the possibility of an iron core, we will also include the relative permeability μ_r of the core in our expression for the magnetic field.

The current through the primary coil is given by Ohm's law:

$$I(t) = \frac{V(t)}{R} = \frac{V_0 \sin(2\pi f t)}{R}.$$

The magnetic field inside the solenoid is:

$$B(t) = \mu_r \mu_0 n_1 I(t) = \frac{\mu_r \mu_0 n_1 V_0 \sin(2\pi f t)}{R}.$$

The magnetic flux through one turn of the coil is:

$$\Phi_B(t) = B(t)A = \frac{\mu_r \mu_0 n_1 A V_0 \sin(2\pi f t)}{R}.$$

Since we are given $n_1 = N_1/\ell$, where N_1 is the number of turns and ℓ is the length of the solenoid, we can rewrite the magnetic flux as:

$$\Phi_B(t) = \frac{\mu_r \mu_0 A N_1 V_0 \sin(2\pi f t)}{\ell R}.$$

2. Use the expression from (1) to derive an expression for the induced emf in the secondary/outer coil.

The secondary coil will experience the same magnetic flux over its N_2 turns, so by Faraday's law, the emf induced in the secondary coil is:

$$\begin{aligned} \mathcal{E}(t) &= N_2 \frac{d\Phi_B}{dt} = N_2 \frac{d}{dt} \frac{\mu_r \mu_0 n_1 A V_0 \sin(\omega t)}{R} = \frac{\mu_r \mu_0 n_1 N_2 A V_0 \omega \cos(\omega t)}{R} \\ &= \frac{\mu_r \mu_0 A N_1 N_2 V_0 2\pi f \cos(2\pi f t)}{\ell R} \\ &= \frac{2\pi \mu_r \mu_0 A N_1 N_2 V_0 f \cos(2\pi f t)}{\ell R}. \end{aligned}$$

This means the amplitude of the induced emf is:

$$\mathcal{E}_{\max} = \frac{2\pi \mu_r \mu_0 A N_1 N_2 V_0 f}{\ell R}.$$

3. Use the expression from (2) to interpret the two graphs you produced in Part 2.

In the first phase, we fixed the amplitude of the primary voltage V_0 while varying the frequency f . From the expression in (2), we see that the amplitude of the induced voltage $\mathcal{E}_{\max}$ is directly proportional to the frequency f , which means the best fit line for the first two graphs should approximate:

$$\text{slope} \approx \frac{2\pi\mu_r\mu_0AN_1N_2V_0}{\ell R}.$$

In the second phase, we fixed the frequency f while varying the amplitude of the primary voltage V_0 . From the expression in (2), we see that the amplitude of the induced voltage $\mathcal{E}_{\max}$ is directly proportional to the amplitude of the primary voltage V_0 , which means the best fit line for the second two graphs should approximate:

$$\text{slope} \approx \frac{2\pi\mu_r\mu_0AN_1N_2f}{\ell R}.$$

Finally, we see that $\mathcal{E}_{\max}$ is also directly proportional to the relative permeability μ_r , which is why the induced voltage is much higher with the iron core than without the iron core.

4. Does the data from your measurements match your theoretical expressions?

For all calculations below, use the following values:

- $A = \pi r_{\text{inner}}^2 = \pi d_{\text{inner}}^2/4 = 2.545 \cdot 10^{-4} \text{ m}^2$
- $N_1 = 150$
- $N_2 = 3300$
- $R = 0.40 \Omega$
- $\ell = 0.11 \text{ m}$

For the first graph, the primary voltage was fixed at $V_0 = 40 \text{ mV}$. Since no iron core was used, we can assume $\mu_r \approx 1$. The theoretical slope is:

$$\text{slope} \approx \frac{2\pi\mu_r\mu_0AN_1N_2V_0}{\ell R} = 0.9041 \text{ mV/Hz}.$$

The slope of the best fit line was approximately 0.9392 mV/Hz. This is a percent error of approximately 3.87%, which means the data matches the theoretical expression in (3) very well.

For the third graph, the frequency was fixed at $f = 500 \text{ Hz}$. Since no iron core was used, we can assume $\mu_r \approx 1$. The theoretical slope is:

$$\text{slope} \approx \frac{2\pi\mu_r\mu_0AN_1N_2f}{\ell R} = 11.30.$$

The slope of the best fit line was approximately 8.50. This is a percent error of approximately 24.75%, which means the data matches the theoretical expression in (3). This is a relatively good match.

For the second and fourth graphs, since we are not given the exact μ_r of the iron core, we cannot calculate a theoretical slope to compare to our data. However, we can see that the slopes of the second and fourth graphs are much higher than the slopes of the first and third graphs, which is consistent with our theoretical expression in (3) since μ_r is much greater than 1 for an iron core.